

Subject : SECP3213 BUSINESS INTELLIGENCE

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Task : Business Intelligence Group Project

Phase : Introduction (Proposal Stage)

Dataset Link : <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce/data>

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1.0 Project Title

Our project title is E-Commerce Performance and Customer Behavior Analytics: A BI Solution for Olist. This project aims to develop a Business Intelligence solution to analyze Olist's e-commerce performance. The analysis focuses on sales performance, customer location, payment behavior, order status, and delivery efficiency using customer, order, and payment datasets.

2.0 Problem Background

Olist is a Brazilian e-commerce platform that connects small and medium-sized businesses to online marketplaces. As an e-commerce platform, Olist manages large amounts of customer, order, payment, and delivery data. However, customer, order, and payment data are often stored separately. This makes it difficult to obtain a complete view of business performance.

The main problem addressed in this project is the lack of an integrated analytical view that combines these datasets for decision-making. This problem is important because e-commerce businesses rely on data to improve sales, customer experience, and logistics operations. Sales analysis helps the business understand revenue trends. Regional analysis helps identify strong and weak markets. Payment analysis helps understand checkout preferences. Delivery analysis helps detect logistics issues that may affect customer satisfaction.

Therefore, this project proposes a BI solution using Alteryx Designer for data preparation and Microsoft Power BI for dashboard development. The final dashboard will help transform raw e-commerce data into meaningful insights for business decision-making.

The expected impact of this project includes better sales performance monitoring, a clearer understanding of customer distribution, and improved payment behavior analysis. Also, it will allow the identification of delivery delays and logistics bottlenecks, and make more effective business decision-making through the use of interactive dashboards.

3.0 Project Objectives

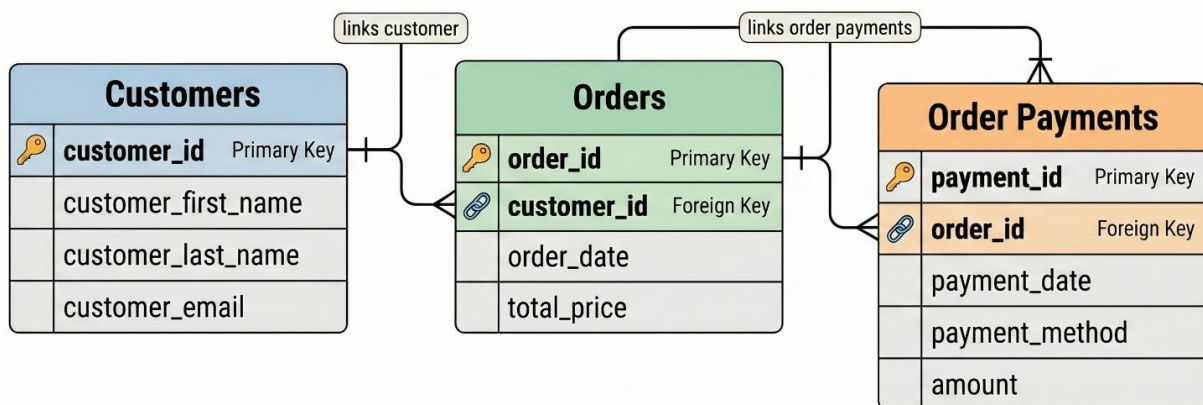
1. To integrate customer, order, and payment datasets into a centralized BI dataset using Alteryx Designer.
2. To analyze e-commerce sales performance based on total revenue, order volume, average order value, and monthly trends.

3. To examine customer payment behavior based on payment method, payment value, and installment usage.
4. To evaluate delivery performance by calculating delivery duration, delivery delay, and delivery status.
5. To develop interactive dashboards using Microsoft Power BI to support business insights.

4.0 Dataset Description

This project uses the Brazilian E-Commerce Public Dataset by Olist, sourced from *Kaggle*. The selected datasets are structured CSV files and are suitable for Business Intelligence analysis because they are related and can be joined using common keys.

Dataset	Description	Main Use
olist_customers_dataset.csv	Contains customer identification and location information	Analyse customer distribution by city and state.
olist_orders_dataset.csv	Contains order status and delivery timeline information.	Analyse order trends, order status, and delivery performance.
olist_order_payments_dataset.csv	Contains payment method, installment, and payment value information.	Analyse revenue and customer payment behavior.



The orders dataset acts as the central dataset because it connects customer information and payment information. The customers dataset will be joined with the orders dataset using `customer_id`, while the payment dataset will be joined with the orders dataset using `order_id`.

One important consideration is that the payments dataset may contain multiple payment records for one order. Therefore, payment data will be summarized by order_id before joining to avoid duplicated revenue values.

These datasets are suitable for the project because they support analysis of sales performance, regional customer behavior, payment preferences, order status, and delivery efficiency. They also meet the project requirement of using at least three relevant datasets.

5.0 Proposed BI Approach

5.1 Tools Selected

5.1.1 Alteryx Designer

Alteryx Designer will be used as the main data preparation and ETL tool. It will be used to import the datasets, clean the data, transform fields, create calculated features, aggregate payment records, join datasets, and export the final analytical dataset.

Alteryx Tool	Purpose
Input Data	Import the CSV datasets.
Select	Check and adjust data types.
Data Cleansing	Handle null values and inconsistent data.
Filter	Filter records based on analysis needs.
Formula	Create calculated fields such as delivery time and delivery delay.
Summarize	Aggregate payment data by order_id.
Join	Combine the three datasets.
Output Data	Export the cleaned dataset for Power BI.

5.1.2 Microsoft Power BI

Microsoft Power BI will be used for data visualization and dashboard development. The cleaned dataset produced from Alteryx will be imported into Power BI to build interactive dashboards. Power BI is suitable because it supports KPI cards, slicers, filters, charts, maps, and business storytelling.

5.2 Data Integration Plan

The data integration process will be performed in Alteryx Designer. First, the three CSV datasets will be imported into Alteryx. The data types and field names will be checked using the Select tool. Next, the payment dataset will be summarized by order_id because one order may contain more than one payment record. After that, the customers dataset will be joined with the orders dataset using customer_id. Finally, the summarized payment dataset will be joined with the order-customer dataset using order_id.

The final integrated dataset will contain customer location, order information, delivery timeline, payment method, and payment value. This dataset will then be exported and used for dashboard development in Power BI.

5.3 Data Cleaning Plan

Data cleaning will be conducted to ensure that the analysis is accurate and reliable. The cleaning steps include checking missing values, duplicate records, invalid payment values, and inconsistent data formats.

Missing delivery dates will be handled carefully. For delivery performance analysis, only delivered orders with valid delivery dates will be used. However, all order statuses will still be retained for order status analysis so that canceled, shipped, unavailable, and processing orders can be analyzed.

Date and time fields will be converted into proper DateTime format to support monthly trend analysis and delivery duration calculations. Location fields such as city and state will also be checked for consistency to ensure accurate regional analysis.

5.4 Data Transformation Plan

Several calculated fields will be created in Alteryx to support deeper analysis.

New Field	Purpose
Order_Year_Month	Used for monthly sales and order trend analysis.
Total_Payment_Value	Used to calculate revenue after payment aggregation.
Delivery_Time_Days	Measures number of days from purchase to delivery.
Delivery_Delay_Days	Measures whether delivery is late compared to estimated date.
Delivery_Status	Classifies delivery as Early, On Time, or Late.

New Field	Purpose
Payment_Category	Classifies payment as Single Payment or Installment Payment.
Order_Value_Category	Groups orders into Low, Medium, or High value.

5.5 Proposed Analysis

Analysis Area	Business Question
Sales Performance	What is the total revenue, total orders, average order value, and monthly sales trend?
Regional Performance	Which states and cities generate the most customers, orders, and revenue?
Payment Behavior	Which payment methods are most used, and how common are installment payments?
Delivery Performance	Which states have the longest delivery time or highest delivery delay?
Order Status	What percentage of orders are delivered, canceled, shipped, or unavailable?

6.0 Expected Business Value

The proposed BI solution will help management monitor sales performance through metrics such as total revenue, total orders, and average order value. This supports better understanding of sales growth and business performance.

Besides, regional analysis will help identify high-performing states and cities. This can support marketing campaigns, regional expansion, and logistics planning. Also, payment analysis will help the company understand customer checkout behavior. If certain payment methods are more popular or generate higher revenue, Olist can improve its payment strategy and customer experience. Apart from that, delivery performance analysis will help identify states with longer delivery times or frequent delays. This can support logistics improvement and help improve customer satisfaction.

In a nutshell, the interactive Power BI dashboard will allow decision-makers to explore the data using filters, slicers, and visual elements. This project will transform raw e-commerce data into meaningful insights that support better decisions in sales, marketing, payment strategy, and delivery operations.